

Who Benefits from Product Changes? Estimating Heterogeneous Treatment Effects in Digital Experiments Using Double Machine Learning

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Abstract

Online controlled experiments are widely used to evaluate product interventions in digital platforms. Traditional A/B testing methods typically focus on estimating average treatment effects, which may obscure important variation in treatment responses across users. This study investigates the use of modern causal inference methods for identifying heterogeneous treatment effects in digital experiments. Using a simulated A/B testing environment, we compare a standard difference-in-means estimator, regression adjustment, and Double Machine Learning (DML) for estimating the effect of a product intervention on user engagement.

The results indicate a positive average treatment effect across all estimation methods, with estimated effects ranging from 4.41 to 4.83 engagement units. However, heterogeneity analysis reveals substantial variation in treatment effects across users. Treatment effects increase systematically with prior engagement, ranging from an average of 3.71 engagement units in the lowest engagement quartile to 6.01 units in the highest quartile. Furthermore, estimated treatment effects closely recover the true treatment effect structure embedded in the simulation, demonstrating the ability of causal machine learning methods to identify meaningful user-level heterogeneity.

These findings highlight the limitations of relying exclusively on average treatment effects and demonstrate the value of combining causal inference and machine learning for digital experimentation. By identifying which users benefit most from an intervention, heterogeneous treatment effect estimation can support more personalized and effective decision-making in technology platforms.

Keywords: Causal Inference, Double Machine Learning, Heterogeneous Treatment Effects, Digital Experiments, A/B Testing, Machine Learning, User Engagement, Personalization

1 Introduction

Online controlled experiments, commonly implemented as A/B tests, have become a cornerstone of decision-making in digital platforms. Technology companies such as Google, Amazon, Meta, and Netflix routinely use randomized experiments to evaluate product features, recommendation systems, interface designs, and marketing interventions (Kohavi et al., 2020). These experiments typically focus on estimating the *average treatment effect* (ATE), which summarizes the overall impact of an intervention across the user population.

Although average treatment effects provide useful information for evaluating interventions, they may conceal important variation in how different users respond to treatment. In many digital environments, users differ substantially in their engagement levels, preferences, demographics, and behavioral patterns. Consequently, a product change that is beneficial on average may have large positive effects for some users and little or no effect for others. Understanding such variation is increasingly important for personalization, targeting, and product optimization.

Recent advances in causal inference and machine learning have introduced methods for estimating *heterogeneous treatment effects* (HTE), which describe how treatment effects vary across individuals or subpopulations. Rather than asking whether an intervention works on average, heterogeneous treatment effect estimation seeks to identify who benefits most from a treatment and under what conditions. This perspective has attracted considerable attention in economics, statistics, medicine, and computer science because of its potential to support more effective and individualized decision-making (Athey and Imbens, 2016; Wager and Athey, 2018).

Among modern approaches for estimating heterogeneous treatment effects, *Double Machine Learning* (DML) provides a flexible framework that combines machine learning with causal inference principles (Chernozhukov et al., 2018). By using orthogonalization and sample splitting, DML reduces bias arising from nuisance parameter estimation while allowing researchers to incorporate high-dimensional covariate information. These properties make DML particularly attractive in digital experimentation settings, where large-scale user-level data are commonly available.

This study investigates the use of causal machine learning methods for identifying heterogeneous treatment effects in digital experiments. Using a simulated A/B testing environment, we compare traditional average-effect approaches with Double Machine Learning and evaluate their ability to recover treatment effect heterogeneity across users. The simulation framework allows the true treatment effect structure to be known, making it possible to assess how accurately heterogeneous effects can be identified from observed data.

This paper makes two primary contributions. First, it demonstrates how reliance on average treatment effects can obscure meaningful variation in user responses to digital interventions. Second, it shows that causal machine learning methods can successfully recover heterogeneous treatment effects and provide insights that are directly relevant to personalization, product experimentation, and decision-making in technology platforms. By linking modern causal inference methods with practical challenges in digital experimentation, this study illustrates the growing role of applied statistics and machine learning in technology-driven decision environments.

2 Literature Review

Online controlled experiments have become one of the most widely used tools for evidence-based decision-making in digital platforms. Technology companies routinely rely on A/B testing to evaluate product features, user interface changes, recommendation systems, and marketing interventions. The popularity of A/B testing stems from its ability to estimate causal effects through randomized assignment while maintaining a relatively simple and interpretable framework for decision-making (Kohavi et al., 2020).

Despite its widespread adoption, traditional A/B testing primarily focuses on estimating average treatment effects (ATE). While average effects provide useful summaries of intervention effectiveness, they may conceal substantial variation in how different users respond to treatment. In modern digital environments, user populations are often highly diverse with respect to demographics, preferences, engagement levels, and behavioral characteristics. As a result, interventions that appear effective on average may affect different users in fundamentally different ways.

Recognizing these limitations, researchers have increasingly focused on the estimation of heterogeneous treatment effects (HTE). Rather than asking whether a treatment works on average, heterogeneous treatment effect analysis seeks to determine for whom a treatment is most effective and under what conditions. This shift from average effects to individualized effects has become particularly important in applications involving personalization, recommendation systems, targeted interventions, and customer engagement strategies.

Early approaches to heterogeneous treatment effect estimation typically relied on subgroup analysis or regression models with interaction terms. Although these methods remain useful in some settings, they require researchers to specify the form of heterogeneity in advance and may perform poorly when treatment effects depend on complex nonlinear relationships among covariates.

Recent developments in causal machine learning have provided more flexible approaches

for uncovering treatment effect heterogeneity. Athey and Imbens (Athey and Imbens, 2016) introduced recursive partitioning methods for identifying subgroups with different causal effects, providing one of the first systematic frameworks for data-driven heterogeneous treatment effect estimation. Building on this work, Wager and Athey (Wager and Athey, 2018) developed causal forests, extending random forest methodologies to estimate heterogeneous treatment effects while enabling valid statistical inference.

Another major advancement in the literature is Double Machine Learning (DML), introduced by Chernozhukov et al. (Chernozhukov et al., 2018). DML combines machine learning algorithms with causal inference principles through orthogonalization and sample splitting procedures that reduce bias arising from nuisance parameter estimation. This framework is particularly attractive in high-dimensional settings where treatment assignment and outcomes may depend on a large number of observed characteristics.

More recent research has extended these methods to improve the estimation of conditional average treatment effects and enhance robustness in practical applications. For example, Semenova and Chernozhukov (2021) propose debiased machine learning procedures for heterogeneous treatment effect estimation, while a growing body of applied research demonstrates the usefulness of causal machine learning methods in healthcare, economics, policy evaluation, and business analytics. These developments suggest that causal machine learning methods have substantial potential for improving decision-making in data-rich environments.

Although the methodological literature on heterogeneous treatment effects has expanded rapidly, comparatively less attention has been given to demonstrating how these methods can be applied within digital experimentation settings that resemble real-world technology platforms. Much of the existing literature focuses on methodological development and theoretical properties, while fewer studies provide practical illustrations of treatment effect recovery in controlled experimentation environments.

This study contributes to the literature by examining the ability of Double Machine Learning to identify heterogeneous treatment effects in a simulated A/B testing environment. Unlike traditional experimentation approaches that emphasize average treatment effects, the analysis focuses on understanding how treatment responses vary across users and whether these differences can be accurately recovered from observed data.

By combining causal inference, machine learning, and digital experimentation, the study provides a practical demonstration of how modern statistical methods can uncover meaningful treatment effect heterogeneity that may be overlooked by average-effect approaches. The findings also highlight the potential value of these methods for personalization, product optimization, and evidence-based decision-making in technology platforms.

3 Data and Simulation Design

This study uses simulated A/B testing data to evaluate the ability of causal machine learning methods to identify heterogeneous treatment effects in digital experiments. Simulation studies are particularly useful for methodological evaluation because the true treatment effect structure is known by design. This allows direct assessment of whether estimation procedures successfully recover underlying treatment effect heterogeneity.

3.1 Experimental Setting

The simulated experiment represents a digital platform evaluating the impact of a new product feature on user engagement. Users are randomly assigned to either a treatment group, which receives access to the new feature, or a control group, which continues using the existing version of the platform.

A total of 5,000 users are generated for the experiment. Treatment assignment is randomized with equal probability, such that each user has a 50% chance of receiving the treatment. Let T_i denote the treatment assignment for user i , where $T_i = 1$ indicates treatment and $T_i = 0$ indicates control.

The primary outcome is user engagement, denoted by Y_i .

3.2 User Characteristics

Each user is assigned a set of observed characteristics intended to resemble variables commonly available in digital experimentation settings.

Specifically, the simulation includes:

- Age, generated from a normal distribution with mean 35 and standard deviation 10;
- Prior engagement, generated from a normal distribution with mean 50 and standard deviation 15;
- Subscription status, generated as a Bernoulli random variable with probability 0.35;
- Mobile device usage, generated as a Bernoulli random variable with probability 0.60.

Let X_i denote the vector of observed user characteristics.

3.3 Outcome Generation Process

The primary outcome is engagement, which is treated as a continuous measure representing overall user activity on the platform.

Baseline engagement is generated according to

$$\mu(X_i) = 20 + 0.40(\text{Prior Engagement}_i) + 3(\text{Subscribed}_i) + 2(\text{Mobile User}_i) - 0.05(\text{Age}_i).$$

This specification reflects the assumption that users with higher prior engagement, active subscriptions, and mobile platform usage tend to exhibit greater baseline activity levels.

3.4 Treatment Effect Heterogeneity

The key feature of the simulation is the presence of heterogeneous treatment effects. The treatment effect is allowed to vary systematically across users according to

$$\tau(X_i) = 2 + 0.06(\text{Prior Engagement}_i) + 1.5(\text{Subscribed}_i) - 0.02(\text{Age}_i).$$

Under this specification, users with higher prior engagement and active subscriptions receive larger benefits from the intervention, while older users experience slightly smaller treatment effects.

The observed outcome is generated as

$$Y_i = \mu(X_i) + \tau(X_i)T_i + \varepsilon_i,$$

where

$$\varepsilon_i \sim N(0, 5^2).$$

This framework generates realistic variation in user responses while ensuring that the true treatment effect structure is known.

3.5 Simulation Objective

The primary objective of the simulation is to evaluate whether modern causal inference methods can identify treatment effect heterogeneity that would be obscured by traditional average-effect approaches. Because the true treatment effect function is known, the simulation provides a controlled environment for assessing the ability of Double Machine Learning

to recover heterogeneous treatment effects across users.

To ensure reproducibility, all simulations were conducted using the R statistical computing environment. A random seed was fixed prior to data generation, and all analyses were performed using the same simulated dataset. This design allows direct comparison of estimation methods under a common experimental framework.

3.6 Descriptive Statistics

Table 1 summarizes the simulated dataset used throughout the analysis. The sample consists of 5,000 users with characteristics designed to resemble a digital experimentation setting. The descriptive statistics confirm that the generated data align with the intended simulation design while exhibiting realistic variation across users.

Table 1: Descriptive Statistics of Simulated Variables

Variable	Mean	SD	Min	Max
Age	34.99	9.95	3.63	69.46
Prior Engagement	49.94	15.04	-7.68	107.72
Subscribed	0.37	0.48	0	1
Mobile User	0.60	0.49	0	1
Treatment	0.48	0.50	0	1
True Treatment Effect	4.85	1.18	1.24	9.08
Engagement	42.74	8.67	14.08	84.53

The average user in the simulated dataset is approximately 35 years old and has a prior engagement score of 49.94. Approximately 36.6% of users are subscribed to the platform, while 60.5% access the platform through a mobile device. Treatment assignment is approximately balanced across groups, reflecting the randomized design of the experiment.

The average true treatment effect is 4.85 engagement units, with substantial variation across users. The observed engagement outcome has a mean of 42.74 and a standard deviation of 8.67, indicating meaningful variation in user activity levels. These characteristics provide a realistic setting for evaluating heterogeneous treatment effect estimation methods.

4 Methods

This section describes the statistical methods used to estimate treatment effects in the simulated digital experiment. Three approaches are considered: a traditional A/B testing estima-

tor, regression adjustment, and Double Machine Learning (DML). Together, these methods illustrate the progression from average treatment effect estimation toward more flexible approaches capable of identifying treatment effect heterogeneity.

4.1 Average Treatment Effect (A/B Testing)

The conventional approach to analyzing randomized experiments is to estimate the average treatment effect (ATE) using the difference in mean outcomes between the treatment and control groups. The estimator is defined as

$$\hat{\tau}_{ATE} = \frac{1}{n_1} \sum_{i:T_i=1} Y_i - \frac{1}{n_0} \sum_{i:T_i=0} Y_i,$$

where n_1 and n_0 denote the number of observations assigned to the treatment and control groups, respectively.

Because treatment assignment is randomized, this estimator provides an unbiased estimate of the average treatment effect. However, it produces only a single population-level estimate and does not reveal variation in treatment effects across users.

4.2 Regression Adjustment

To improve estimation precision, a regression-adjusted model is estimated using observed user characteristics:

$$Y_i = \alpha + \tau T_i + X_i' \beta + \varepsilon_i,$$

where X_i represents the vector of observed covariates.

The coefficient τ measures the average treatment effect after controlling for differences in user characteristics. Although regression adjustment often improves efficiency, it generally assumes a linear functional form and may fail to capture complex treatment effect heterogeneity unless interaction terms are explicitly incorporated into the model.

4.3 Double Machine Learning

To estimate treatment effects while flexibly accounting for high-dimensional covariate information, this study employs Double Machine Learning (DML), introduced by Chernozhukov et al. (2018). DML combines machine learning algorithms with causal inference principles to reduce bias arising from nuisance parameter estimation.

The DML framework relies on estimating two nuisance functions:

$$m(X_i) = \mathbb{E}[Y_i | X_i],$$

which represents the conditional expectation of the outcome, and

$$e(X_i) = \mathbb{E}[T_i | X_i],$$

which represents the propensity score or treatment assignment model.

In this study, both nuisance functions are estimated using random forest learners. Random forests are well suited for capturing nonlinear relationships and complex interactions among user characteristics without requiring the researcher to specify these relationships in advance.

The key idea of DML is to construct orthogonalized residuals by removing the influence of the nuisance functions:

$$\tilde{Y}_i = Y_i - \hat{m}(X_i),$$

$$\tilde{T}_i = T_i - \hat{e}(X_i).$$

The treatment effect is then estimated using the residualized outcome and treatment variables:

$$\hat{\tau}_{DML} = \frac{\sum_i \tilde{T}_i \tilde{Y}_i}{\sum_i \tilde{T}_i^2}.$$

To reduce overfitting, DML employs sample splitting and cross-fitting procedures in which nuisance functions are estimated on one subset of the data and evaluated on another. The analysis was conducted in the R statistical computing environment using the `DoubleML` package (Bach et al., 2024). Random forest learners from the `ranger` package were used to estimate both the outcome model and the treatment assignment model. Following common practice in the causal machine learning literature, five-fold cross-fitting was employed to reduce overfitting and improve the robustness of treatment effect estimation. These procedures improve robustness and allow machine learning methods to be incorporated while maintaining valid statistical inference.

4.4 Assessment of Treatment Effect Heterogeneity

The primary objective of this study is not only to estimate the average treatment effect but also to investigate whether treatment effects vary systematically across users.

To evaluate heterogeneity, users are grouped according to their prior engagement levels, and treatment effects are examined across engagement quartiles. This approach allows assessment of whether highly engaged users benefit differently from the intervention compared with less engaged users.

Because the true treatment effect function is known in the simulation, estimated treatment effects can be compared directly with the true treatment effects embedded in the data-generating process. This comparison provides a direct measure of how well treatment effect heterogeneity is recovered.

4.5 Summary of Empirical Strategy

The empirical analysis proceeds in four stages:

1. Estimate the average treatment effect using a traditional A/B test.
2. Estimate a regression-adjusted treatment effect using observed covariates.
3. Apply Double Machine Learning with random forest learners to estimate the treatment effect.
4. Evaluate treatment effect heterogeneity by comparing estimated effects across engagement groups and assessing recovery of the true treatment effect structure.

Together, these methods provide a framework for comparing traditional average-effect estimators with modern causal machine learning approaches designed to uncover user-level treatment effect heterogeneity.

5 Results

This section presents the results from the simulated digital experiment. We begin by reporting the average treatment effect using a standard A/B test, followed by regression-adjusted estimates and Double Machine Learning estimates. We then examine heterogeneous treatment effects and evaluate the ability of the estimation procedure to recover the true treatment effect structure embedded in the simulation.

5.1 Average Treatment Effect

We first estimate the average treatment effect (ATE) using the difference in mean engagement between the treatment and control groups. The estimated ATE is 4.41 engagement units,

indicating that users who received the new feature exhibited higher engagement on average than users in the control group.

Although this estimate suggests that the intervention is effective overall, it provides only a population-level summary and does not reveal whether the treatment effect differs across users.

5.2 Regression-Adjusted Results

Next, we estimate a regression-adjusted model controlling for age, prior engagement, subscription status, and mobile device usage. The estimated treatment effect is 4.81 engagement units ($p < 0.001$), indicating a positive and statistically significant effect of the intervention after adjusting for observed user characteristics.

Several covariates are also significantly associated with engagement. Prior engagement exhibits a strong positive relationship with the outcome, while age has a small negative association. Subscription status and mobile device usage are both positively associated with engagement.

The similarity between the A/B test estimate and the regression-adjusted estimate is expected because treatment assignment was randomized. Nevertheless, the regression model improves precision by accounting for variation explained by the observed covariates.

5.3 Double Machine Learning Results

We then apply Double Machine Learning (DML) using random forest learners for both the outcome and treatment assignment models. The DML estimate of the treatment effect is 4.83 engagement units with a standard error of 0.157 ($p < 0.001$).

The DML estimate is highly consistent with both the A/B test and regression-adjusted estimates. This agreement provides evidence that the simulated treatment effect is being recovered accurately across multiple estimation approaches.

While the average treatment effect remains positive across all methods, the primary objective of this study is to understand whether treatment effects vary across users rather than focusing exclusively on the average effect.

Table 2: Comparison of Treatment Effect Estimates

Method	Estimated Treatment Effect
A/B Test	4.41
Regression Adjustment	4.81
Double Machine Learning	4.83

5.4 Heterogeneity Analysis

The simulation was designed to generate heterogeneous treatment effects as a function of user characteristics. Specifically, treatment effects increase with prior engagement and subscription status while decreasing slightly with age.

Figure 1 displays the distribution of the true heterogeneous treatment effects used in the data-generating process. The distribution ranges approximately from 1 to 9 engagement units, with most users experiencing treatment effects between 4 and 6 units. This demonstrates that the intervention does not affect all users equally.

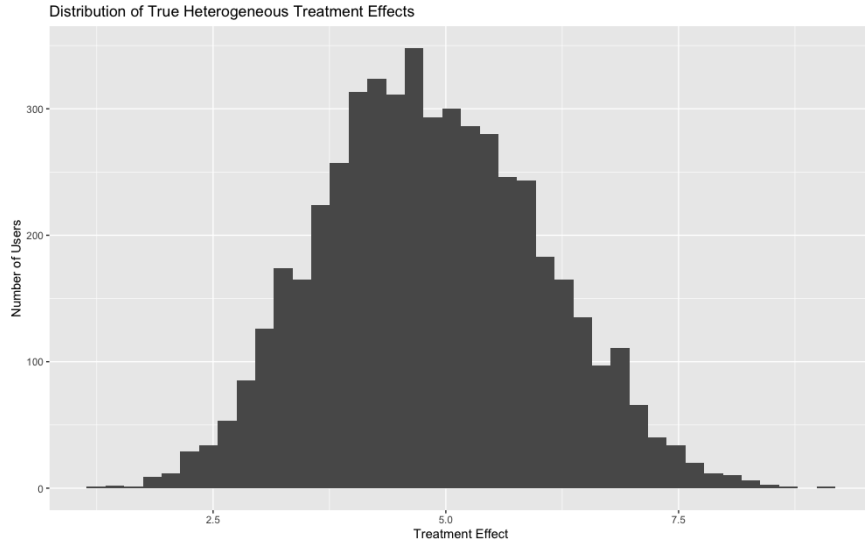


Figure 1: Distribution of True Heterogeneous Treatment Effects

Figure 2 illustrates the relationship between prior engagement and the treatment effect. Users with higher baseline engagement tend to experience larger gains from the intervention, whereas users with lower baseline engagement experience smaller treatment effects.

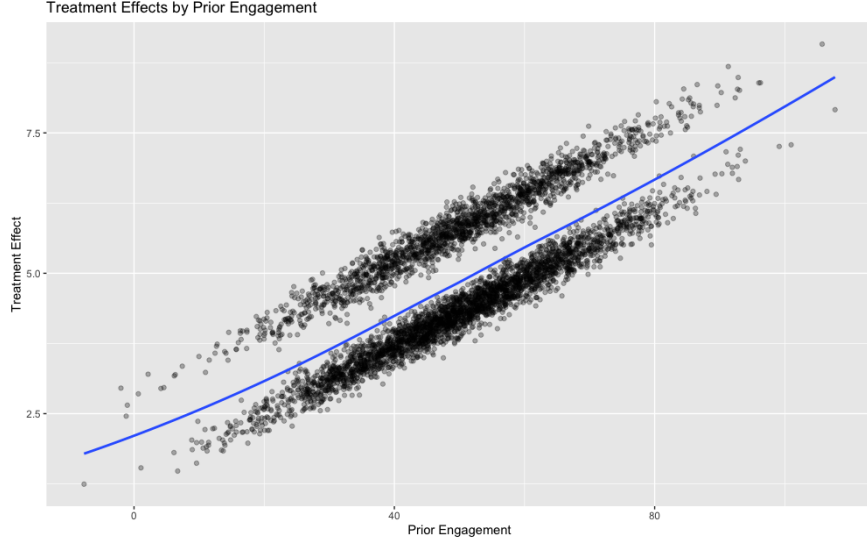


Figure 2: Treatment Effects by Prior Engagement Level

The positive relationship observed in Figure 2 suggests that the effectiveness of the feature is concentrated among users who are already active on the platform. Consequently, a single average treatment effect masks substantial variation in treatment response across the user population.

5.5 Recovering Heterogeneous Treatment Effects

To further examine treatment effect heterogeneity, users were divided into quartiles based on prior engagement. The average treatment effect was then calculated within each engagement group using the true treatment effect from the simulation design.

Table 3 shows that treatment effects increase monotonically across engagement groups. Users in the lowest engagement quartile have an average treatment effect of 3.71 engagement units, while users in the highest engagement quartile have an average treatment effect of 6.01 engagement units.

Table 3: Average Treatment Effects by Prior Engagement Group

Engagement Group	Mean Prior Engagement	Mean Treatment Effect
1	30.8	3.71
2	45.1	4.53
3	54.7	5.14
4	69.1	6.01

Figure 3 visualizes this pattern. The upward trend indicates that users with higher prior engagement benefit more from the product intervention.

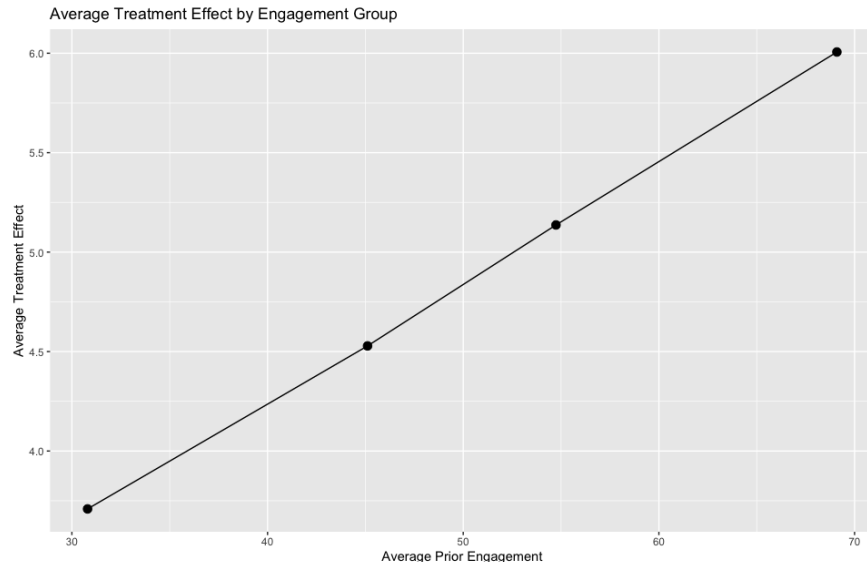


Figure 3: Average Treatment Effect by Prior Engagement Group

These results demonstrate that the treatment effect is not constant across users. Instead, the effect of the product intervention is strongest among users who were already more engaged before treatment. This supports the central argument of the study: relying only on the average treatment effect can obscure important user-level differences that are relevant for product decision-making.

5.6 Recovery Analysis

A key advantage of simulation studies is that the true treatment effect is known. This allows us to evaluate whether the estimated treatment effects successfully recover the underlying heterogeneity embedded in the data-generating process.

Users were divided into quartiles based on prior engagement, and treatment effects were estimated within each group. Table 4 compares the true treatment effects used in the simulation with the estimated treatment effects obtained from the observed data.

Table 4: Recovery of Heterogeneous Treatment Effects

Engagement Group	True Effect	Estimated Effect	Absolute Error
1	3.71	3.61	0.10
2	4.53	4.26	0.27
3	5.14	5.29	0.15
4	6.01	5.42	0.59

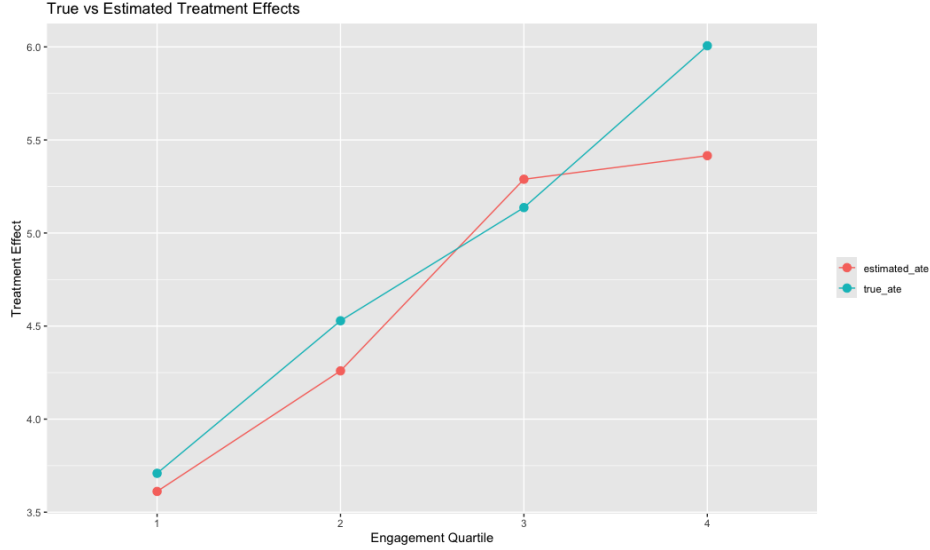


Figure 4: True versus Estimated Treatment Effects Across Engagement Groups

Figure 4 provides a visual comparison between the true treatment effects used in the simulation and the treatment effects estimated from the observed data. The estimated effects closely follow the true effects across all engagement groups, indicating that the underlying heterogeneity is successfully recovered.

The estimated treatment effects closely track the true treatment effects across all engagement groups. Although some estimation error is present, the overall pattern of heterogeneity is recovered successfully. In particular, users with higher prior engagement consistently exhibit larger treatment effects than users with lower prior engagement.

These findings demonstrate that treatment effect heterogeneity can be identified from observed data and that the estimated effects closely approximate the underlying treatment effect structure. This result supports the use of causal machine learning methods for uncovering user-level heterogeneity in digital experiments.

5.7 Summary of Findings

Across all estimation methods, the treatment was found to have a positive effect on user engagement. The traditional A/B test estimated an average treatment effect of 4.41 engagement units, while regression adjustment and Double Machine Learning produced estimates of 4.81 and 4.83 engagement units, respectively. The consistency of these estimates provides evidence that the intervention generated a meaningful increase in engagement.

Beyond average effects, the analysis revealed substantial treatment effect heterogeneity. Users with higher levels of prior engagement consistently experienced larger treatment effects than less engaged users. Treatment effects increased from 3.71 engagement units in the lowest engagement quartile to 6.01 engagement units in the highest engagement quartile.

The recovery analysis further demonstrated that the estimated treatment effects closely approximated the true treatment effects embedded in the simulation. Although some estimation error was present, the overall heterogeneity pattern was successfully recovered. These findings suggest that modern causal inference methods can provide insights beyond average treatment effects and identify meaningful variation in treatment responses across users.

6 Discussion

This study examined the use of causal machine learning methods for identifying heterogeneous treatment effects in digital experiments. Although all three estimation approaches—the traditional A/B test, regression adjustment, and Double Machine Learning—identified a positive treatment effect, the results demonstrated that the average treatment effect alone provides an incomplete picture of intervention effectiveness.

A central finding of this study is that meaningful treatment effect heterogeneity existed across users. While the average treatment effect was approximately 4.8 engagement units, individual treatment effects varied considerably depending on user characteristics. In particular, users with higher levels of prior engagement consistently experienced larger treatment effects than users with lower engagement levels. These findings illustrate how average treatment effects can mask important differences in treatment response across a population.

The recovery analysis provides additional support for the usefulness of causal machine learning methods. Because the true treatment effect structure was known, it was possible to directly evaluate the accuracy of the estimated treatment effects. The results showed that the estimated effects closely tracked the true effects across engagement groups, indicating that the underlying heterogeneity was successfully recovered. This finding suggests that modern causal inference methods can provide meaningful insights into treatment effect variation

rather than merely estimating average effects.

The practical implications of these findings are particularly relevant for digital experimentation. Technology companies routinely conduct large-scale online experiments to evaluate new product features, recommendation systems, user interface changes, and marketing interventions. Traditional experimentation frameworks typically focus on determining whether a treatment improves outcomes on average. However, understanding which users benefit most from an intervention may be equally important for decision-making.

For example, a recommendation algorithm may substantially increase engagement among highly active users while having little effect on less active users. Similarly, a new platform feature may improve retention among subscribers but generate minimal benefits for casual users. In such settings, heterogeneous treatment effect estimation can support more effective targeting, personalization, and resource allocation strategies. Rather than deploying interventions uniformly across all users, organizations may use treatment effect estimates to identify groups that are most likely to benefit.

From a methodological perspective, this study highlights the growing integration of causal inference and machine learning within applied statistics. Traditional statistical methods remain essential for estimating average treatment effects and evaluating intervention effectiveness. However, modern causal machine learning approaches extend these methods by enabling the analysis of complex and potentially nonlinear treatment effect heterogeneity. As digital platforms continue to generate increasingly large and complex datasets, the ability to combine statistical inference with machine learning will become increasingly important.

This study also illustrates how methodologies developed in fields such as economics, public health, and policy evaluation can be adapted to digital experimentation. The same causal inference principles used to evaluate public policies and healthcare interventions can be applied to questions involving user engagement, recommendation systems, and product optimization. This connection highlights the broad applicability of causal inference methods across diverse domains.

6.1 Potential Applications in Digital Platforms

Although this study relies on simulated data, the analytical framework is directly applicable to a wide range of real-world digital experimentation settings. Technology companies routinely collect large-scale user-level data that can be used to evaluate heterogeneous treatment effects and support personalized decision-making.

Potential applications include recommendation systems, online advertising experiments, customer retention interventions, subscription optimization, and user engagement initiatives.

For example, a streaming platform may use heterogeneous treatment effect estimation to identify which users are most likely to respond to a new recommendation algorithm, while an e-commerce platform may use similar methods to determine which customers are most responsive to targeted promotions.

The framework may also be applied to product experimentation environments in which organizations seek to understand not only whether an intervention is effective on average, but also which users derive the greatest benefit. By identifying treatment effect heterogeneity, companies can allocate resources more efficiently, improve personalization strategies, and enhance user experiences through targeted interventions.

These applications highlight the growing importance of causal machine learning methods in technology-driven decision environments and demonstrate the potential value of integrating heterogeneous treatment effect estimation into modern experimentation workflows.

Several limitations should be acknowledged. First, the analysis was conducted using simulated data in which the true treatment effect structure was specified by design. Although simulation studies provide an important framework for methodological evaluation, real-world digital experiments often involve more complex behavioral processes, measurement error, and unobserved factors. Second, the simulation considered a relatively small set of user characteristics. In practice, technology platforms may collect hundreds or thousands of potential predictors, creating additional challenges and opportunities for treatment effect estimation. Third, the analysis focused exclusively on engagement as the outcome of interest. Future studies could investigate outcomes such as click-through rates, conversion rates, retention, customer lifetime value, or revenue generation.

Future research should extend this framework to real-world digital experimentation datasets. Applying causal machine learning methods to observational and experimental data from technology platforms would provide a valuable opportunity to evaluate treatment effect heterogeneity in practical settings. Additional work could also compare Double Machine Learning with alternative heterogeneous treatment effect estimators, including causal forests, Bayesian causal models, and meta-learning approaches.

Overall, the findings demonstrate that modern causal inference methods can uncover meaningful treatment effect heterogeneity that is not captured by traditional average-effect approaches. By moving beyond average treatment effects, causal machine learning provides a richer understanding of user behavior and offers valuable opportunities for improving personalization, experimentation, and decision-making in digital environments.

7 Conclusion

This study investigated the use of modern causal inference methods for identifying treatment effect heterogeneity in digital experiments. Using a simulated A/B testing environment, we compared traditional average treatment effect estimation with machine learning-based causal inference approaches, with particular emphasis on understanding how treatment effects vary across users.

The results showed that the intervention generated a positive average treatment effect across all estimation methods, including standard A/B testing, regression adjustment, and Double Machine Learning. While these approaches consistently indicated that the intervention increased user engagement, the heterogeneity analysis revealed substantial variation in treatment responses. Users with higher levels of prior engagement experienced larger treatment effects than less engaged users, demonstrating that average effects alone do not fully capture the impact of an intervention.

A major contribution of this study was the evaluation of treatment effect recovery. Because the true treatment effect structure was known, it was possible to directly assess the accuracy of the estimated effects. The results demonstrated that the estimated treatment effects closely tracked the true treatment effects across engagement groups, providing evidence that modern causal machine learning methods can successfully identify meaningful variation in treatment responses.

These findings underscore the limitations of relying exclusively on average treatment effects in digital experimentation. Although average effects remain useful for evaluating overall intervention performance, they may obscure important differences across users. By identifying which users benefit most from an intervention, causal machine learning methods provide information that can support more targeted, personalized, and effective decision-making.

Beyond digital experimentation, this study highlights the growing convergence of applied statistics, causal inference, and machine learning. As organizations increasingly rely on large-scale data to guide decisions, methods capable of uncovering treatment effect heterogeneity will become increasingly important for personalization, product optimization, and resource allocation.

Future research should extend this framework to real-world digital experimentation datasets and explore additional outcomes such as click-through rates, retention, conversion, customer lifetime value, and revenue. Applying these methods to large-scale experimental and observational datasets would provide further insight into the practical value of causal machine learning in technology-driven environments.

In conclusion, this study demonstrates that modern causal inference methods can move beyond average treatment effects and provide a richer understanding of user behavior. By uncovering who benefits most from an intervention, causal machine learning offers a powerful framework for improving experimentation, personalization, and evidence-based decision-making in digital platforms.

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